VISVESVARAYA TECHNOLOGICAL UNIVERSITY, BELAGAVI

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
SCHEME OF TEACHING AND EXAMINATION (CREDIT BASED SYSTEM)

Course Title: Advanced Data Structures and Algorithms

Course Code: 21CS42
CIE Marks: 50
SEE Marks: 50
Total Hours: 50
Credits: 04

COURSE OBJECTIVES:

1. Understand the design principles and analysis of complex data structures.
2. Formulate and solve real-world problems using graph algorithms and dynamic programming.
3. Evaluate algorithm complexity using asymptotic notations.

MODULE 1: Linear & Hierarchical Structures

Review of Stacks, Queues, Circular Queues, Priority Queues. Binary Search Trees, AVL Trees, 2-3 Trees, Red-Black Trees, B-Trees and B+ Trees: Operations, Insertion, Deletion, and Amortized Analysis.

MODULE 2: Advanced Graph Algorithms & Flow Networks

Topological sorting, Strongly Connected Components, Minimum Spanning Trees (Prim's and Kruskal's), Single Source Shortest Path (Dijkstra and Bellman-Ford), All-Pairs Shortest Path (Floyd-Warshall), Maximum Flow: Ford-Fulkerson algorithm.

MODULE 3: Disjoint Sets & Dynamic Programming

Disjoint set data structures, Union by rank and path compression. Dynamic programming paradigm: Matrix chain multiplication, Longest common subsequence, 0/1 Knapsack problem, Optimal binary search trees.

MODULE 4: String Matching & Randomized Algorithms

Naive string matching, Rabin-Karp algorithm, Knuth-Morris-Pratt (KMP) algorithm, Boyer-Moore algorithm, Suffix trees. Randomized quicksort, Primality testing.

MODULE 5: NP-Completeness & Approximation Algorithms

Class P, NP, NP-Hard, and NP-Complete problems. Cook's Theorem statement, Reducibility proofs: Vertex Cover, Clique, Subset Sum, Hamiltonian Cycle. Approximation algorithms for Vertex Cover and Traveling Salesperson Problem.

TEXT BOOKS:

1. Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein, "Introduction to Algorithms", 3rd Edition, PHI Learning / MIT Press, 2009, ISBN: 978-0262033848.

2. Mark Allen Weiss, "Data Structures and Algorithm Analysis in C++", 4th Edition, Pearson Education, 2014, ISBN: 978-0132847377.
3. Ellis Horowitz, Sartaj Sahni, Sanguthevar Rajasekaran, "Fundamentals of Computer Algorithms", 2nd Edition, Universities Press, 2008, ISBN: 978-8173716126.

REFERENCE BOOKS:

1. Michael T. Goodrich, Roberto Tamassia, Michael H. Goldwasser, "Data Structures and Algorithms in Python", 1st Edition, John Wiley & Sons, 2013, ISBN: 978-1118290279.
2. Robert Sedgewick, Kevin Wayne, "Algorithms", 4th Edition, Addison-Wesley Professional, 2011, ISBN: 978-0321573513.
3. Anany Levitin, "Introduction to the Design and Analysis of Algorithms", 3rd Edition, Pearson, 2012, ISBN: 978-0132316811.
4. Alfred V. Aho, John E. Hopcroft, Jeffrey D. Ullman, "Data Structures and Algorithms", 1st Edition, Pearson Education, 1983, ISBN: 978-0201000238.

COURSE OUTCOMES:

Upon successful completion of the course, students will be able to:
CO1: Analyze worst-case and amortized running times of complex algorithms.
CO2: Implement advanced trees and graph topologies for high-throughput computing.
CO3: Apply dynamic programming strategies to optimization challenges.